

Naive Bayes Foundations: Conditional Probability

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Leads to:** Independent and mutually exclusive events ([Note 83](#)); Bayes' theorem ([Note 85](#)).

1. Overview

Key point: The probability of A given B, $P(A | B)$, is the probability of A once we know B has happened. $P(A | B)$ equals $P(A \cap B) / P(B)$.

The next Notes build the **Naive Bayes** classifier, a very fast classification algorithm that works well on text, famously for document classification and spam filtering (scikit-learn user guide §1.9). It rests on a few ideas from probability: conditional probability, independent events, and Bayes' theorem. This Note covers the first.

Conditional probability answers questions of the form “how likely is A, now that we know B is true?”. Conditional probability is used throughout probability and machine learning, and Bayes' theorem is built directly on it.

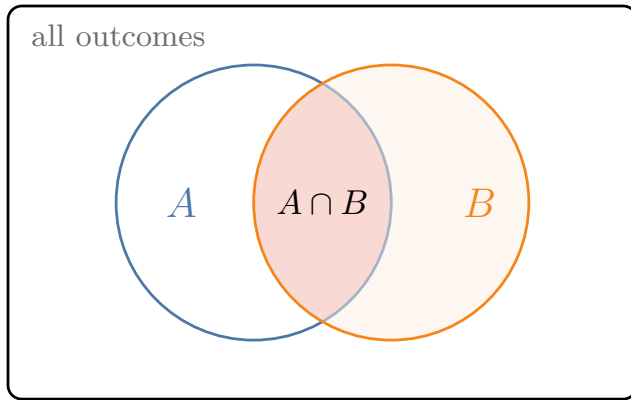
2. The definition

Key point: $P(A | B) = P(A \cap B) / P(B)$, as long as $P(B)$ is not zero.

For two events A and B:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

- $P(A | B)$ is read “the probability of A given B”.
- $A \cap B$ (A intersection B) means “both A and B happen”.



$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

once B is known to have happened,
 B becomes the whole world:
 what share of B is also A ?

Figure 1: Conditional probability as a share of B

The idea (Figure 1): once we know B has happened, every outcome outside B is ruled out. B becomes the whole world, and the question is what share of it also belongs to A .

An everyday picture: a friend draws a card from a deck and says “it is red”. We stop thinking about all 52 cards and look only at the 26 red ones. The chance that the card is a heart is now 13 out of 26, one half, instead of one quarter.

3. An example with two dice

Key point: Rolling two dice has 36 equally likely outcomes. Conditioning on “sum ≤ 10 ” shrinks the space to 33 outcomes.

Roll two dice together. Each outcome is a pair, (die 1, die 2): $(1, 1), (1, 2), \dots, (6, 6)$. There are $6 \times 6 = 36$ outcomes, all equally likely. This set of all outcomes is the **sample space**.

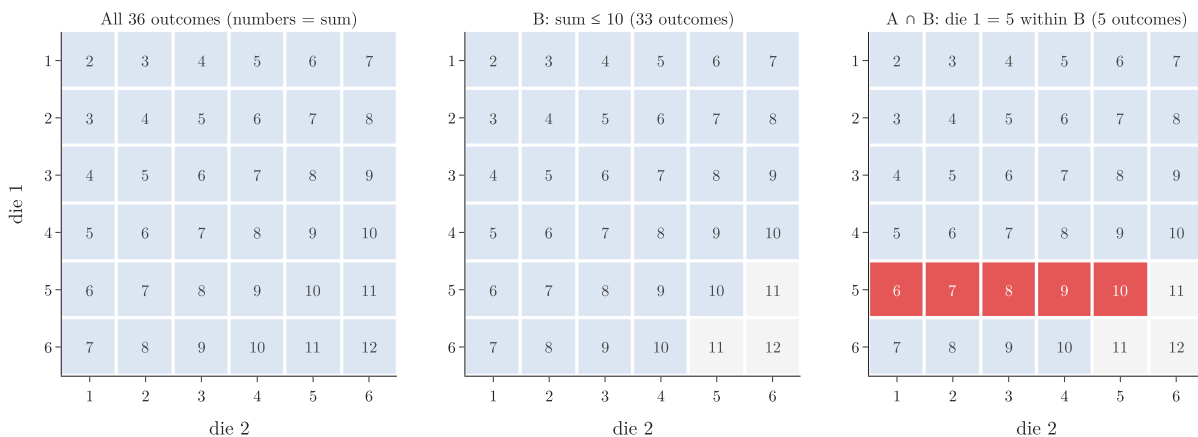


Figure 2: The 36 outcomes, the event B , and $A \cap B$

Figure 2 lays the sample space out as a grid, with the sum of the two dice in each cell.

3.1 Two ordinary probabilities

Key point: $P(\text{die 1} = 5) = 6/36$; $P(\text{sum} \leq 10) = 33/36$.

With equally likely outcomes, a probability is the number of favourable outcomes divided by the total.

- **A: die 1 shows 5.** Event A is the whole fifth row of the grid: 6 outcomes. $P(A) = 6/36 = 1/6$.
- **B: the sum is at most 10.** Only three outcomes break it: (5, 6), (6, 5) and (6, 6), with sums 11, 11 and 12. So $P(B) = 33/36 = 11/12$ (Figure 2, middle).

3.2 A conditional probability by counting

Key point: Count only inside B: 5 of its 33 outcomes have die 1 = 5, so $P(A | B) = 5/33$.

Now ask: **what is the probability that die 1 shows 5, given that the sum is at most 10?**

Since B is known to have happened, only its 33 outcomes are possible. These 33 outcomes form the **reduced sample space**. Inside it, die 1 shows 5 in the outcomes (5, 1), (5, 2), (5, 3), (5, 4) and (5, 5); the sixth, (5, 6), has sum 11 and is ruled out (Figure 2, right). So

$$P(A | B) = \frac{5}{33} \approx 0.152$$

Knowing B has lowered the chance of A slightly, from $6/36 \approx 0.167$, because B ruled out one of A's outcomes.

Figure 3 shows the shrinking as motion, after the geometric picture of Bayes' rule by Sanderson (3Blue1Brown). Watch the three cells outside B disappear; the 33 cells left are then laid out again to fill the same area as the original 36, because B is now the whole world. A's 5 cells (red) take $5/33$ of it. The last scene swaps the roles, as in section 4.

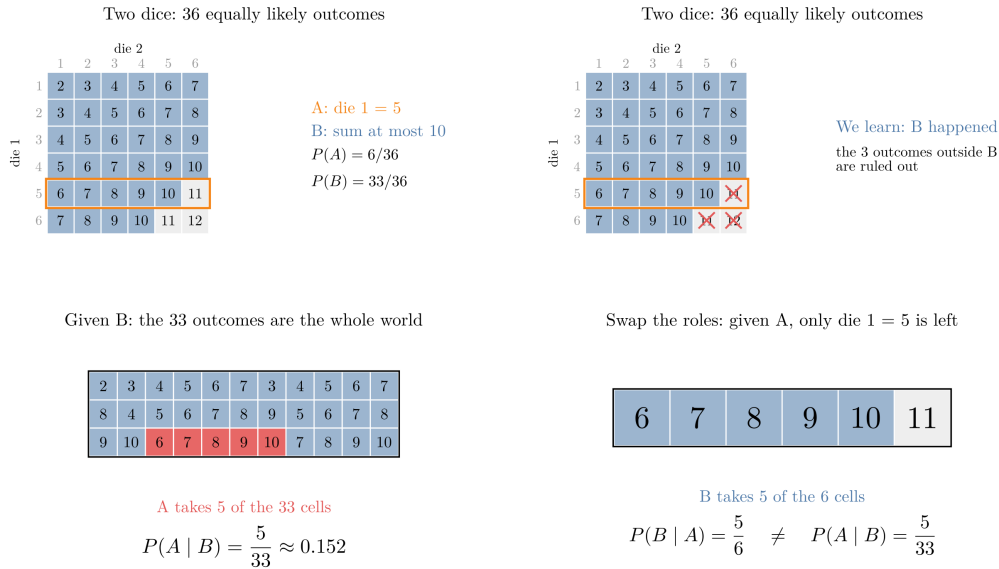


Figure 3: Conditioning shrinks the sample space: the 3 outcomes outside B are ruled out, the 33 left become the whole world and A covers 5 of them; given A instead, B covers 5 of A's 6 outcomes

3.3 The same answer from the formula

Key point: $P(A \cap B) = 5/36$ and $P(B) = 33/36$; dividing gives $5/33$.

Using the formula instead, with the full sample space of 36:

- $P(A \cap B)$: die 1 is 5 **and** the sum is at most 10. These are the same 5 outcomes, so $P(A \cap B) = 5/36$.
- $P(B) = 33/36$.

$$P(A | B) = \frac{5/36}{33/36} = \frac{5}{33}$$

The 36s cancel, leaving exactly the count in the reduced sample space. The formula and the counting say the same thing.

Python: Counting all 36 outcomes.

```
from fractions import Fraction
from itertools import product
```

```
outcomes = list(product(range(1, 7), repeat=2))
B = [o for o in outcomes if sum(o) <= 10]
A_and_B = [o for o in B if o[0] == 5]
Fraction(len(A_and_B), len(B))
```

```
# 33 outcomes
# 5 outcomes
# 5/33
```

A simulation of a million rolls gives 0.1512, close to $5/33 = 0.1515$.

4. $P(A \mid B)$ is not $P(B \mid A)$

Key point: Swapping the roles changes the answer: $P(\text{sum} \leq 10 \mid \text{die 1} = 5) = 5/6$, not $5/33$.

The order matters. Given that die 1 shows 5, the reduced space is its 6 outcomes, of which 5 have a sum of at most 10:

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{5/36}{6/36} = \frac{5}{6}$$

So $P(B \mid A) = 5/6$ while $P(A \mid B) = 5/33$ (Figure 3, last scene). Confusing the two is a classic mistake. Bayes' theorem (two Notes later) is exactly the rule that converts one into the other.

5. Summary

- Conditional probability: $P(A | B) = P(A \cap B) / P(B)$.
- Knowing B shrinks the sample space to B; count A's share inside it.
- Two dice: $P(\text{die 1} = 5 | \text{sum} \leq 10) = 5/33$, against $P(\text{sum} \leq 10 | \text{die 1} = 5) = 5/6$.

6. Sources

Built from

- CampusX, “Naive Bayes Classifier | Part 1 | Conditional Probability”, YouTube, <https://www.youtube.com/watch?v=Ty7knppVo9E>
- **Sanderson, G. (3Blue1Brown)**: “Bayes’ theorem”, 3blue1brown.com/lessons/bayes-theorem. The picture of conditioning as shrinking the space of possibilities; Figure 3 recreates it with our own dice example.

Other references

- **scikit-learn user guide**: Section 1.9, “Naive Bayes”, scikit-learn 1.9.

7. Key terms

Term	Meaning
Sample space	The set of all possible outcomes of an experiment
Event	A set of outcomes, such as “the sum is at most 10”
Intersection ($A \cap B$)	The event that both A and B happen
Conditional probability	The probability of an event given that another event has happened: $P(A B)$
Reduced sample space	The outcomes that remain possible once the condition is known
Naive Bayes	A classification algorithm based on Bayes’ theorem (later Notes)